Assignment 1

1. Create table Student (Rno, Name, DOB, Gender, Class, College, City, Marks)

2. Insert 5 records in student table

3. Display the information of all the students

4. Display the detail structure of student table

5. Display Rno, Name and Class information of 'Bhubaneswar' students.

6. Display information on ascending order of marks

7. Change the marks of Rno 5 to 89.

8. Change the name and city of Rno 9.

9. Delete the information of 'Kolkata' city records

10. Delete the records of student where marks<30

11. Create table emp which has the following attributes (employee table) (empno, ename, job, sal, deptno)

12. Insert appropriate records in above tables.

13. Get employee no and employee name who works in dept no 10

14. Display the employee names of those clerks whose salary > 2000

15. Display name and sal of Salesperson & Clerks

16. Display all details of employees whose salary between 2000 and 3000

17. Display all details of employees whose dept no is 10, 20, or 30

18. Display name of those employees whose commission is NULL

19. Display dept no & salary in ascending order of dept no and with in each dept no salary should be in descending order

20. Display name of employees that starts with 'C'

21. Display name of employees that ends with with 'C'

22. Display name of employees having two 'a' or 'A' chars in the name

23. Display the name of the employees whose second char is ‘b’ or ‘B’

24. Display the name of the employees whose first or last char is ‘a’ or ‘A’

Database Systems Lab Assignment 2

Objective: To learn basics SQL DDL and DML statements

1. Create the tables with following database schema for a company database:

   Employee (eno, ename, deptno, salary, sex, DoB)

   Department (dno, dname, dloc, dmgrno)

   Eno and dno are primary keys of Employee and Department tables respectively. Deptno is a foreign key of Employee table referring dno of Department table. Dmgrno is a foreign key of Department table referring eno of Employee table. Salary can't be less than 10k.

2. Insert at least 10 records in Employee table and 3 records in department tables.
3. Write the following SQL queries
   (a) List the details of all employees belonging to a given department number (deptno will be taken from user)
   (b) List all employees whose age below 30 years
   (c) List the salary of all managers of different departments along with their names.
   (d) List the department name and their manager names.
   (e) List the name of all employees in a given department whose salary is more than 100k (deptno will be taken from uses)
   (f) Find the name of employees whose salary is maximum.
   (g) Increase the salary of all employees in a given department by 10% (deptno will be taken from user).
   (h) List the name of employees whose name contains string "TH"
   (i) Find the total salary of all female employees of those department which are located in Bangalore

Assignment 3

Create the company database with the following relational schema and instance. Create relevant primary and foreign keys.

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

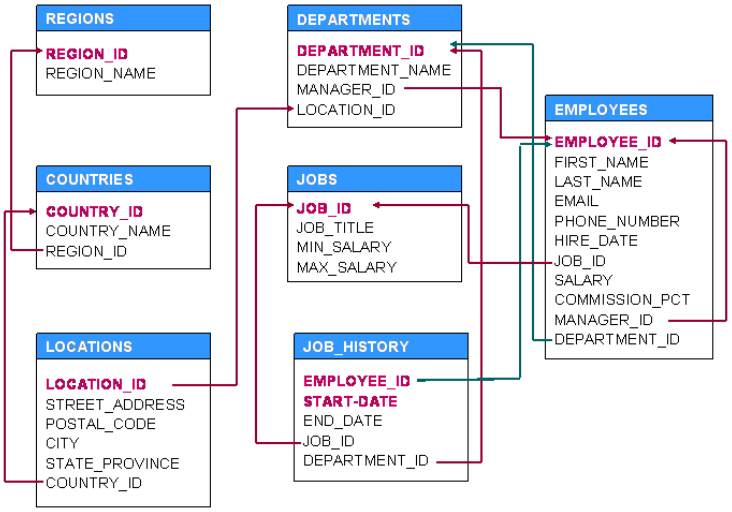
Write SQL Statements for the following queries.

a. For every project located in ‘Stafford’, list the project name, the controlling department name, and the department manager’s name, address, and birth date.
b. If every employee working on the ‘ProductX’ project is given a 10 percent raise, show the resulting salaries along with names of the employees.
c. Retrieve a list of employees and the projects they are working on, ordered by department and, within each department, ordered alphabetically by last name, then first name.
d. Retrieve the names of all employees and their direct managers in department 5 who work more than 10 hours per week on the ProductX project.
e. Retrieve the Essns of all employees who work the same (project, hours) combination on some project that employee ‘John Smith’ (whose Ssn =‘123456789’) works on. [Use nested query structure]
f. Retrieve the name of each employee who has a dependent with the same first name and is the same sex as the employee. [Use nested query structure]
g. Find the sum of the salaries of all employees of the ‘Research’ department, as well as the maximum salary, the minimum salary, and the average salary in this department.
h. For each department that has more than or equal to 3 employees, retrieve the department number and the number of its employees who are making more than $30,000.
i. Transfer those employees to department located in Stafford whose date of birth is in 1965.
j. List all the employee names who works in all the projects in which Smith works.
k. Add a new attribute called “Commission” in the Employee Table. Update the commission value for each employees. Keep some null values for some employees as they don’t possess any commission. Now display, employee name and their total salary.
l. Delete Newbenefits project details from PROJECT relation. This must automatically delete referenced tuples from other relations.
m. Display Age and Salary of all employees in a given deptname. The deptname must be taken as user-input.

Assignment 4

Consider the given HR database and write SQL programs for the following queries.

1. Display the EMPNAME as FIRSTNAME||LASTNAME, MONTHLY PAY as SALAY, ANNUAL PAY, DNAME for each employees in descending order of salary
2. Display total number of employees and name of dept manager for a given deptname. Deptname will be taken as user input.
3. Display the EMPNAME as FIRSTNAME||LASTNAME, MONTHLY PAY as SALAY, ANNUAL PAY, DNAME for each employees in descending order of salary including commission percentage
4. Display the EMPNAME as FIRSTNAME||LASTNAME, EXPERIENCE (IN YEARS) for each employees in descending order of EXPERIENCES
   [Hint: EXTRACT (YEARS FROM SYSDATE) gives year]
5. Display OLD-SALARY, NEW-SALARY of employees after updating the salary of each employee in deptno 90 by 10%
6. Display deptno, deptname and Total_salary of those departments which spend >100000 for salary
7. Create a view showing maximum and minimum salary earners name for each job ID
   Insert an employee record with higher salaries and check the consistency of the view
8. Create a view containing employee ID, Deptname, city and country of each employees
9. Display name and salary of the manager who has a team size>5.
10. Retrieve all employees who are working in department 10 and who earn at least as much as any (i.e., at least one) employee working in department 30.
11. Create a view contains the name, job title, commission percent and the annual salary of employees working in the department 20. Increase the salary of the employees by 10% through EMPLOYEES table and observe the details from the view. Update the commission percent of job type CLERK with increase in 10%. Observe the change in annual salary through view.



Assignment 5

Table 1: SalesPeople

Snum is Primary key

Sname is Unique constraint

Snum	Sname	City	Comm
1001	Peel	London	.12
1002	Serres	Sanjose	.13
1004	Motika	Landon	.11
1007	Rifkin	Barcelona	.15
1003	Axelrod	Newyork	.10

Table 2: Customers

Cnum is Primary

Key

City has not null constraint .

Snum is foreign key constraint refers Snum column of SalesPeople table.

Cnum	Cname	City	Snum
2001	Hoffman	London	1001
2002	Giovanni	Rome	1003
2003	Liu	Sanjose	1002
2004	Grass	Berlin	1002
2006	Clemens	London	1001
2008	Cisneros	Sanjose	1007

2007	Pereira	Rome	1004

<u>Table 3:</u> Orders

Onum is Primary

key

Cnum is foreign key refers to Cnum column of Customers table.

Snum is foreign key refers Snum column of SalesPeople table.

Onum	Amt	Odate	Cnum	Snum
3001	18.69	3-10-1990	2008	1007
3003	767.19	3-10-1990	2001	1001
3002	1900.10	3-10-1990	2007	1004
3005	5160.45	3-10-1990	2003	1002
3006	1098.16	3-10-1990	2008	1007
3009	1713.23	4-10-1990	2002	1003
3007	75.75	4-10-1990	2004	1002
3008	4273.00	5-10-1990	2006	1001
3010	1309.95	6-10-1990	2004	1002
3011	9891.88	6-10-1990	2006	1001

1) Count the number of Salesperson whose name begin with **‘a’/’A’**.

2) Display all the Salesperson whose all orders worth is more than Rs. 2000.

3) Count the number of Salesperson belonging to **Newyork**.

4) Display the number of Salespeople in London and **Paris**.

5) Display the number of orders taken by each Salesperson and their date of orders.

6) Write a query that counts the number of Salespeople registering orders each day.

7) Write a query that selects the first customer in alphabetical order, whose name begins with **‘G’**.

8) Find out the largest orders for Snum 1002 & 1007.

9) Find out the maximum single order amount of a Salesperson over Rs. 3000 in a day.

10) Find out the no. of Salesperson who belongs to same city and have same commission percentage.

11) Find those salesperson name who live in any one of the city of customers (do it both with sub-query and join)

12) Find those salesperson name,customers name who belong to any one of the cty of customers
(do it both with sub-query and join)

13) Find those salesperson name who belong to the city of their customer (do it both with sub-query and join)

14) Find those salesperson name who belong to the city of their customer (do it with co-related sub-query)

15) Find those salesperson name,customer name where salesperson is assigned/not assigned to any customer

16) Find those customer name who is not assigned to any salesperson

17) Find the highest order of each salesperson

18) Find the names of salesperson and their highest order

19) Find those orders of salesperson which is more than his average orders

20) List those salesperson who has more than two customers.